

No. of Functional Features Included in the Solution

Date	03 July 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Multi-Model Emotion Classification	Evaluates student study challenges across a 12-emotion distribution spectrum using a custom BiLSTM network and a DistilBERT transformer framework.	src/emotion_pipeline.py	Done	High
2	In-Memory Config Patching Layer	Automated zip-architecture patching layer that scrubs out breaking serialized configuration parameters from .keras model files at execution runtime.	src/emotion_pipeline.py	Done	High
3	Keyword Heuristic Boosting Layer	Deterministically scales and overrides classification layers using text keywords mapping directly to domain-specific emotional strain signals.	src/keyword_rules.py	Done	Medium
4	GenAI Tailored Guidance Integration	Communicates with the Google Gemini API to return context-aware study roadmaps matching the student's emotional state and field.	src/gemini_guidance.py	Done	High
5	Interactive Analytics Dashboard	Visualizes text data interactively using historical distribution charts, time-series usage lines, and error-proof log tracking modules.	app.py & src/logger.py	Done	High

Feature Summary:

Metric	Count / Value
Total Features Planned	5
Total Features Implemented	5
Core / Must-Have Features	3 (Dual-Model Pipelines, Config Repair, GenAI Guidance Integration)
Additional / Nice-to-Have Features	2 (Keyword Rule Boosting Layers, Multi-Tab Analytics Dashboard)
Features Tested & Verified	5

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	Multi-tab interactive user layout built using the Streamlit layout component framework.	Text intake panels, model comparison radios, and dynamically sizing Plotly chart frames.
2	Backend / Logic	Tokenization, structural text array padding routines, and neural network inference pathways.	texts_to_sequences mapping layouts, tensor generation, and probability Softmax layers.
3	Database / Storage	Local flat-file tabular persistence log architecture tracking execution timestamps and system outputs.	Transaction row logging, log clearing metrics, and Pandas tracking reads.
4	API / Integration	Generative LLM broker pipeline facilitating external API orchestration layer executions.	Token usage handlers, custom prompt instruction wrappers, and template fallback backups.
5	Security / Authentication	Input data preparation boundaries, variable shielding, and source file environment isolation.	Text cleansing via regex utilities, system environment variables, and active .gitignore configurations.